

Mahindra University Hyderabad
École Centrale School of Engineering
Minor I Examinations

Program: B. Tech. Branch: CM Year: II Semester: II
Subject: Stochastic Processes (MA2213)

Date: 26/02/2026
Time Duration: 1.5 Hours

Start Time: 02.00 PM
Max. Marks: 30

Instructions:

1. All questions are compulsory.
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Q 1:

Marks: 7

- a) Let $Z(t) = X(t) - 2X(t-1)$, where $X(t)$ is a Wiener process with parameter $\sigma^2 = 0.25$.
- (i) Find the pdf of $Z(t)$.
 - (ii) Find mean and auto-covariance functions of $Z(t)$.
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Q 2:

Marks: 7

- a) Let $Y(t) = X(t+s) - \beta X(t)$, where $X(t)$ is a WSS random process.
- (i) Determine whether $Y(t)$ is also a WSS random process.
 - (ii) Find the cross-covariance function of $Y(t)$ and $X(t)$.
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Q 3:

Marks: 8

- a) Consider the moving average process: $Y_n = (2/3)X_n + (1/3)X_{n-1}$, $X_0 = 0$. Find the mean, variance, and auto-covariance of Y_n if :
- (i) X_n is a zero-mean step process.
 - (ii) X_n is a Bernoulli process with $p = 1/2$.
- Determine whether Y_n is a WSS process.
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Q 4:

Marks: 8

- a) Let $N_1(t)$ be a Poisson process with arrival rate λ_1 that is started at $t = 0$. Let $N_2(t)$ be another Poisson process that is independent of $N_1(t)$, that has arrival rate λ_2 , and that is started at $t = 1$. Then, find the probability mass function of $N(t) = N_1(t) + N_2(t)$.
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